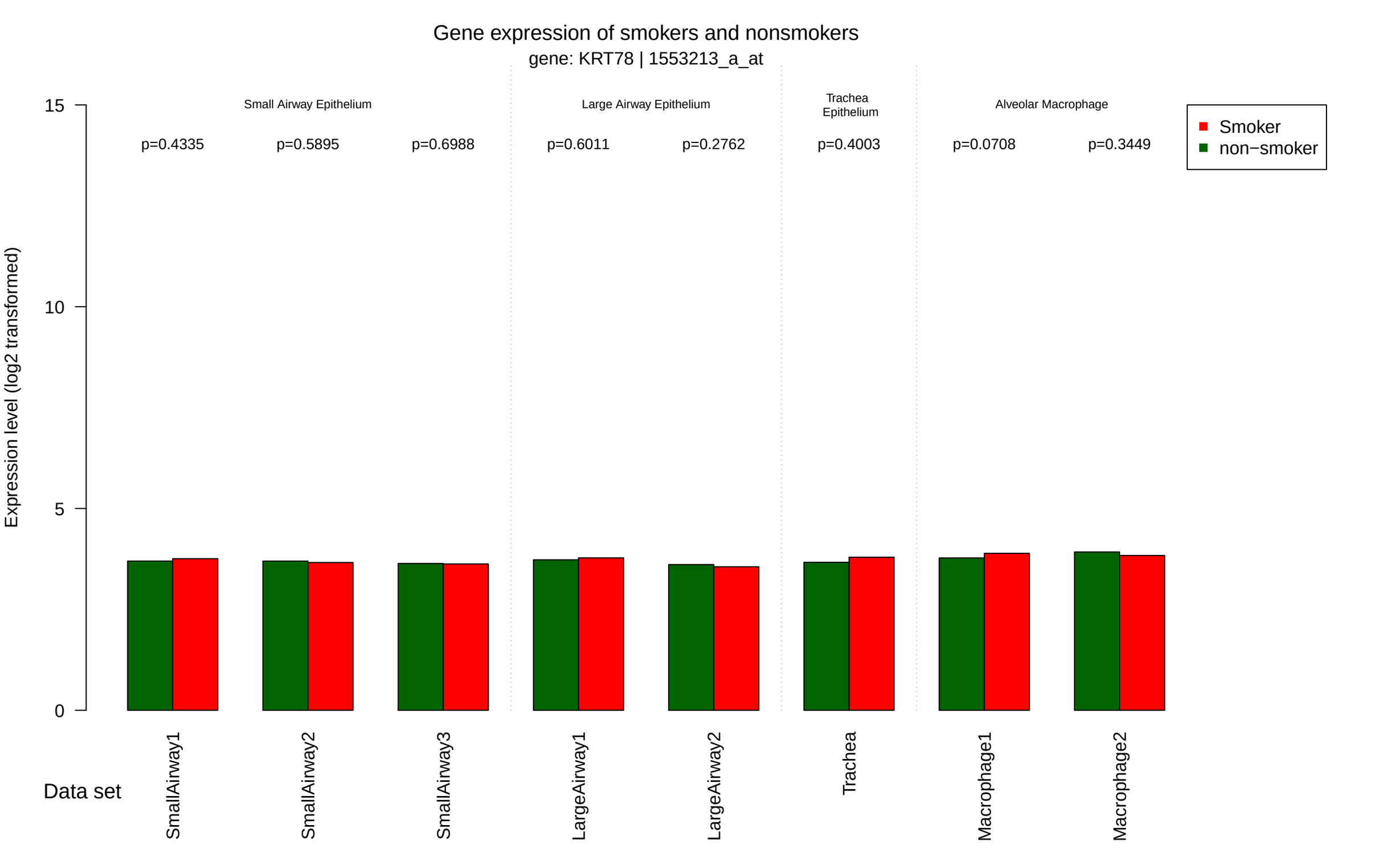
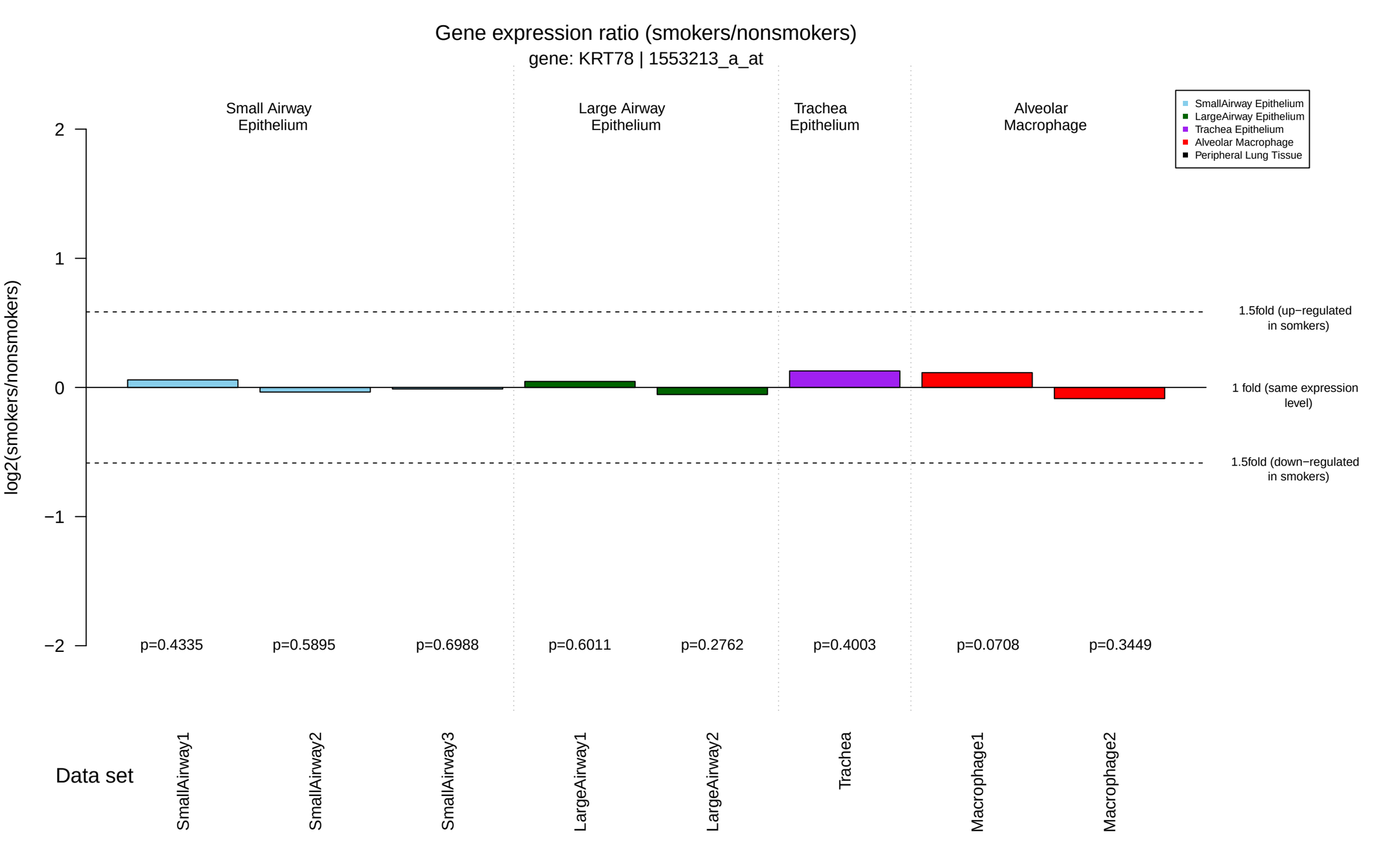
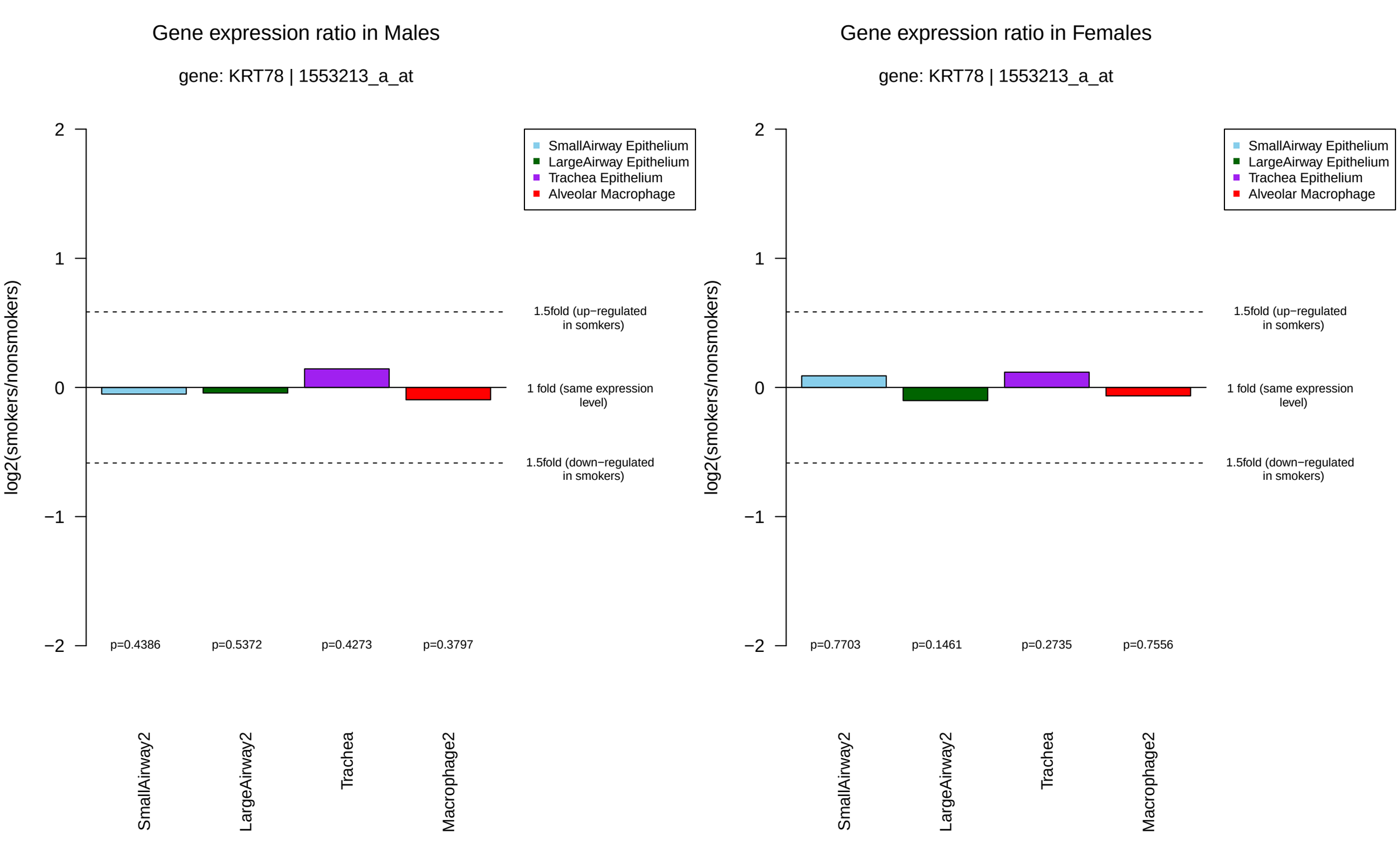
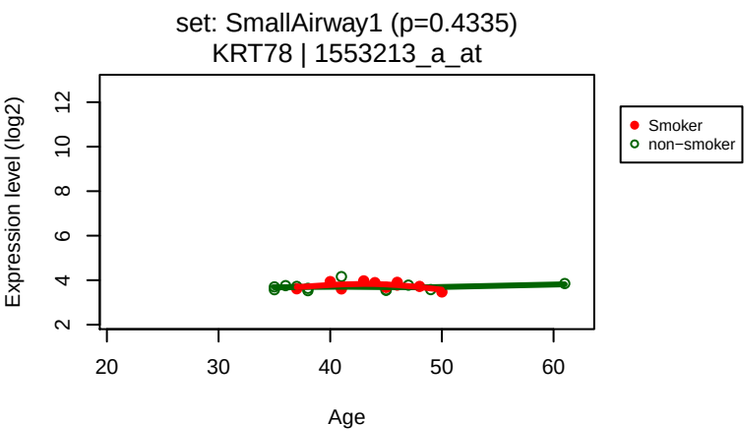
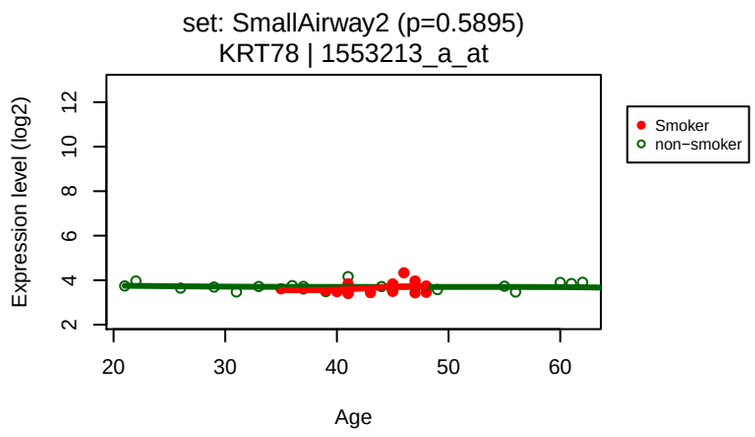
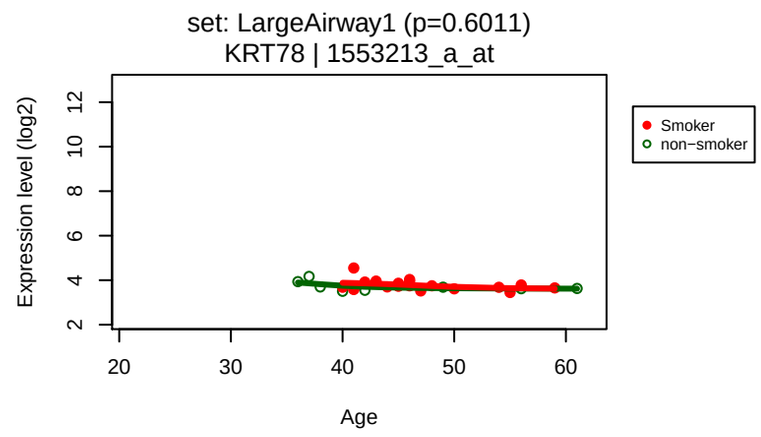

Gene expression of smokers and nonsmokers
gene: KRT78 | 1553213_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4335
p=0.5895
p=0.6988
p=0.6011
p=0.2762
p=0.4003
p=0.0708
p=0.3449
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

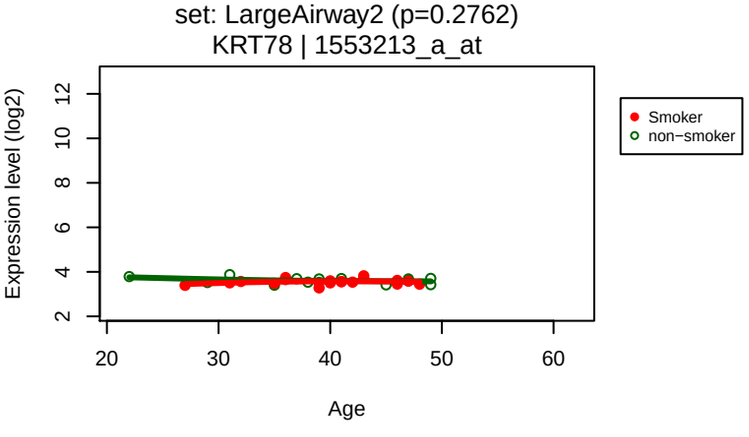

Gene expression ratio (smokers/nonsmokers)
gene: KRT78 | 1553213_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4335
p=0.5895
p=0.6988
p=0.6011
p=0.2762
p=0.4003
p=0.0708
p=0.3449
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

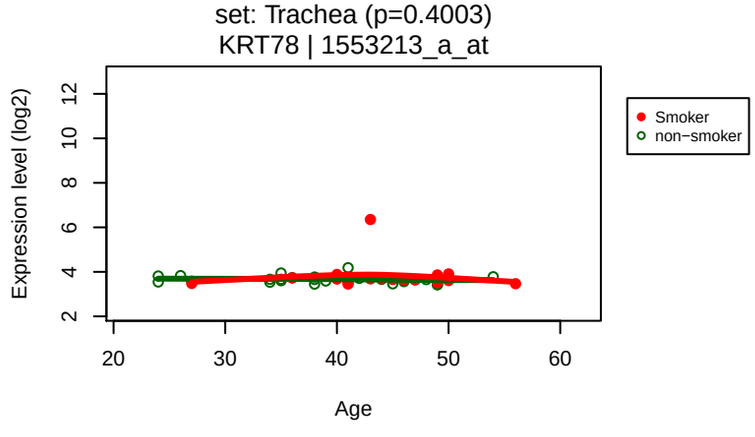

Gene expression ratio in Males
gene: KRT78 | 1553213_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4386
p=0.5372
p=0.4273
p=0.3797
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRT78 | 1553213_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7703
p=0.1461
p=0.2735
p=0.7556
SmallAirway2
LargeAirway2
Trachea
Macrophage2

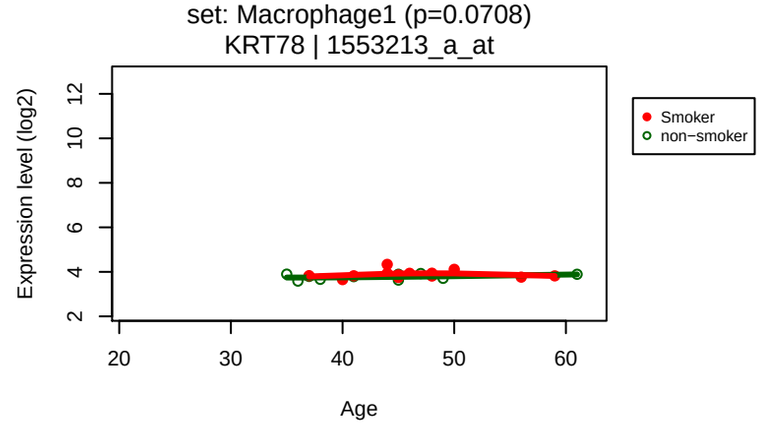

set: SmallAirway1 (p=0.4335)
KRT78 | 1553213_a_at
Expression level (log2)
Age
Smoker
non−smoker

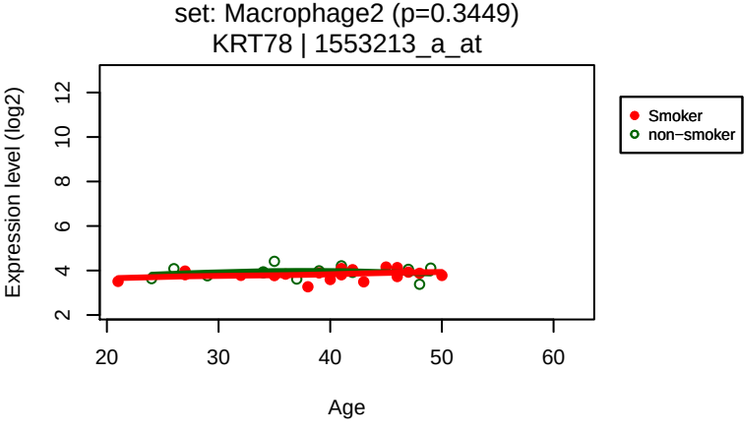

set: SmallAirway2 (p=0.5895)
KRT78 | 1553213_a_at
Expression level (log2)
Age
Smoker
non−smoker

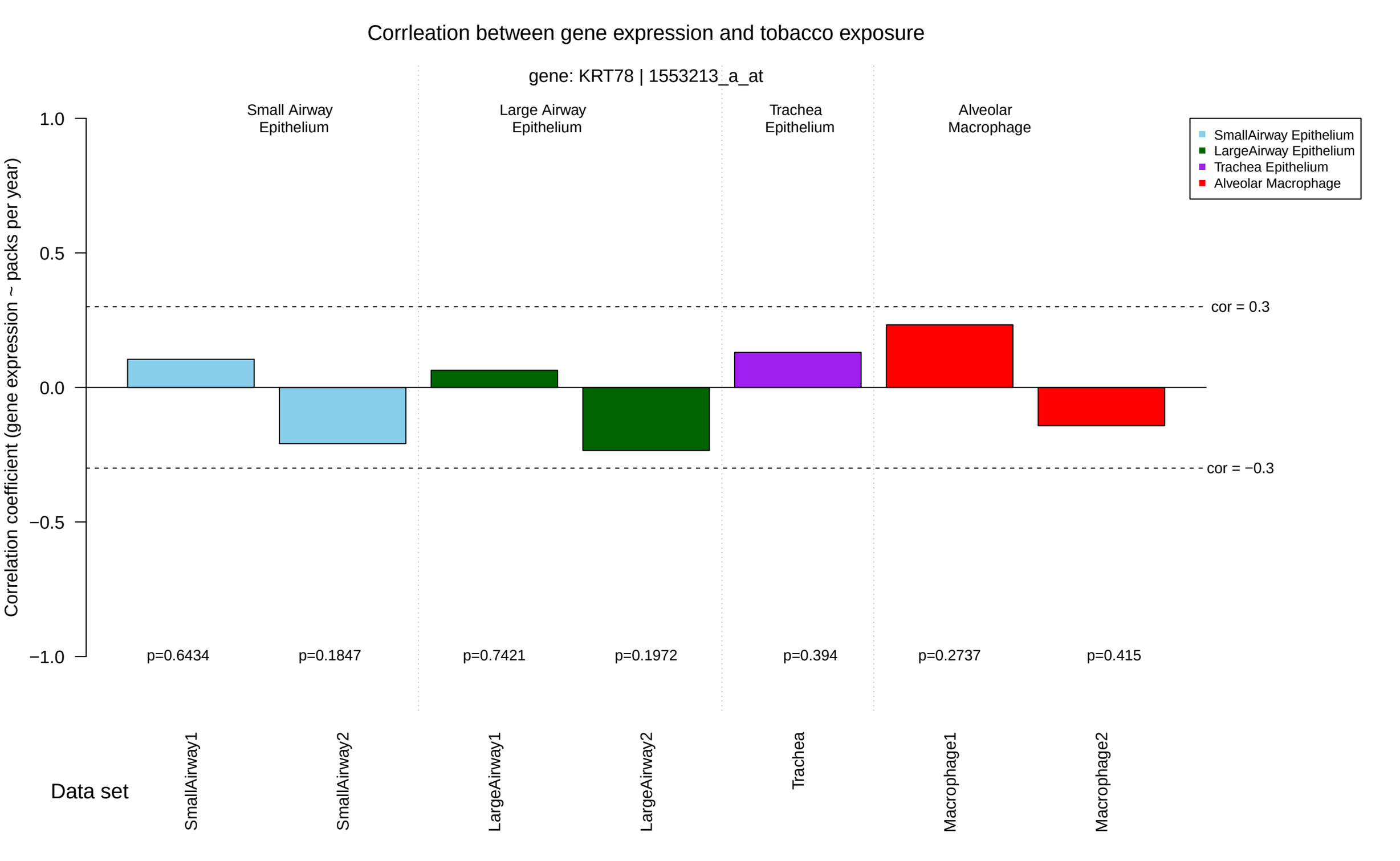

set: LargeAirway1 (p=0.6011)
KRT78 | 1553213_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2762)
KRT78 | 1553213_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4003)
KRT78 | 1553213_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0708)
KRT78 | 1553213_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3449)
KRT78 | 1553213_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRT78 | 1553213_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6434
p=0.1847
p=0.7421
p=0.1972
p=0.394
p=0.2737
p=0.415
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2